

Exercise Sheet 2

MATH 598 / 784 • High-Dimensional Probability

For discussion Friday, September 18, 2026

1. A lower bound for the Gaussian maximum

Let Z be a standard normal random variable, and let $G = (G_1, \dots, G_d)$ have independent standard normal coordinates.

1. Use the Paley–Zygmund inequality (or Cauchy–Schwarz) to show that, for every $t \geq 0$,

$$\mathbb{P}\{\|G\|_\infty \geq t\} \geq \frac{d \mathbb{P}\{|Z| \geq t\}}{1 + (d-1) \mathbb{P}\{|Z| \geq t\}}.$$

2. Deduce that, as $d \rightarrow \infty$,

$$\mathbb{E}\|G\|_\infty \geq \sqrt{2 \log d} - o(1).$$

2. Gaussian width of the ℓ^p -ball

For a fixed $1 < p < \infty$, let $B_p^d = \{x \in \mathbb{R}^d : \sum_{i=1}^d |x_i|^p \leq 1\}$. Recall that its Gaussian width is

$$w(B_p^d) = \mathbb{E} \sup_{x \in B_p^d} \langle G, x \rangle,$$

where G has independent standard normal coordinates. Compute this width up to a vanishing relative error: find an explicit function $f_p(d)$ such that

$$\frac{w(B_p^d)}{f_p(d)} \rightarrow 1 \quad \text{as } d \rightarrow \infty.$$

3. Small ball probabilities (Exercise 2.2.10)

Let $X_1, \dots, X_N$ be independent non-negative random variables, each with a probability density bounded above by 1.

1. Show that, for each i ,

$$\mathbb{E} \exp(-tX_i) \leq \frac{1}{t} \quad \text{for all } t > 0.$$

2. Deduce that, for every $\varepsilon > 0$,

$$\mathbb{P}\left\{\sum_{i=1}^N X_i \leq \varepsilon N\right\} \leq (e\varepsilon)^N.$$

Hint. Rewrite the event as $\sum_{i=1}^N (-X_i/\varepsilon) \geq -N$ and use the exponential Markov inequality, independence, and part 1.

4. Chernoff's inequality: small deviations (Exercise 2.3.5)

Let $X_1, \dots, X_N$ be independent Bernoulli random variables with $\mathbb{P}\{X_i = 1\} = p_i$. Set $S_N = \sum_{i=1}^N X_i$ and $\mu = \mathbb{E}S_N = \sum_{i=1}^N p_i$. Show that, for every $\delta \in (0, 1]$,

$$\mathbb{P}\{|S_N - \mu| \geq \delta\mu\} \leq 2 \exp(-c\mu\delta^2),$$

where $c > 0$ is an absolute constant.

Hint. For $\mu > 0$, Chernoff's upper- and lower-tail bounds are $\mathbb{P}\{S_N \geq t\} \leq e^{-\mu}(e\mu/t)^t$ for $t > \mu$ and $\mathbb{P}\{S_N \leq t\} \leq e^{-\mu}(e\mu/t)^t$ for $0 < t < \mu$ (Theorem 2.3.1 and Exercise 2.3.2). Apply these with $t = (1 \pm \delta)\mu$ and analyze the exponents. Treat the endpoint $\delta = 1$ by a limit in the lower-tail bound, and handle $\mu = 0$ separately.

Problems 3 and 4 are adapted from Roman Vershynin, *High-Dimensional Probability: An Introduction with Applications in Data Science*, first edition, Cambridge University Press, 2018, Exercises 2.2.10 and 2.3.5. The author provides the [first-edition PDF](#) free of charge. Exercise numbering differs in the second edition.